

January 26, 2026

Why AI Is Running Into Silver

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Memory, Electricity, and the Physics of Exponential Growth

Every period of exponential innovation carries a familiar illusion: that progress is limited primarily by imagination, software, or intelligence. In the early phases of a technological wave, this illusion often holds. Capability compounds rapidly, costs fall, and constraints appear to dissolve.

History shows, however, that exponential systems do not escape physics. They collide with it.

When that collision occurs, progress does not slow because ideas run out, but because bottlenecks emerge in places that were previously invisible. This is why my 2026 outlook paper was titled “The Physical World Upgrade: Why AI’s Next Chapter Demands a Hardware Renaissance.” Artificial intelligence has now reached that point.

Much of the public debate around AI remains centered on models, algorithms, and compute. The conversation oscillates between optimism about intelligence gains and anxiety over job displacement, regulation, or software valuations. Markets ask when software bottoms will form or when certain data-center trades will work again. Meanwhile, engineers and governments are signaling a different concern entirely.

Two inputs, seemingly unrelated, have begun behaving in strikingly similar ways: high-bandwidth memory in the digital world, and silver in the physical one.

Since September, both DRAM prices and silver prices have moved sharply higher. Both are essential to the physical-world upgrade required for AI’s next phase. At Davos, Jensen Huang estimated that this global upgrade could reach \$85 trillion over the next 15 years, far larger than the \$5–8 trillion currently discussed for data centers alone. The real investment wave is not confined to training models or scaling centralized compute, but to rebuilding the infrastructure that allows intelligence to operate everywhere.

Constraint Migration: From Software to Hardware to Materials

AI does not just create abundance in software; it exposes scarcity in the physical world.

DDR3, older-generation DRAM, shows what happens when production exits too early. Silver shows what happens when strategic importance is misunderstood for too long. Both have experienced sharp repricing, supply stress, and rising strategic attention. Both are small in mass, expensive relative to alternatives, and essential to system functionality.

Most importantly, both reveal the same structural truth: exponential systems collapse onto their narrowest constraint.

To understand why silver now matters in the age of AI, it helps to begin with memory.

Memory as the First Hidden Bottleneck

In the first two years following the launch of ChatGPT, compute appeared to be the dominant constraint. Progress was measured through GPU shipments, capital expenditures, and the assumption that more arithmetic power would directly

translate into more intelligence.

As models scaled, a different reality emerged. Compute without memory stalls. Arithmetic units increasingly sat idle while waiting for data. Latency, bandwidth, and proximity became more important than raw processing power. Intelligence, it turned out, was not just about calculation, it was about recall, context, and state.

High-bandwidth memory became a choke point not because it was conceptually exciting, but because it sat at the boundary between potential and realization. Without sufficient memory bandwidth, AI systems could not utilize the compute they already had. The constraint was not intelligence itself, but access to intelligence. Silver will prove to be the electrical analog, the interface material that determines whether power infrastructure can keep pace with the intelligence it must serve.

As this bottleneck became visible, memory shifted from an afterthought to a strategic asset.

Inference, the Edge, and Data Movement

This shift becomes even more pronounced as AI moves from centralized training toward distributed inference and edge deployment away from the cloud and closer to the physical world it must sense, interpret, and act upon. Inference is fundamentally a data-movement problem. Performance depends less on raw computation and more on how quickly data can be retrieved, moved, and acted upon under tight energy and latency constraints.

Leaving the cloud changes the nature of the system. In centralized environments, latency can be masked and inefficiencies tolerated. At the edge, distance becomes unforgiving. Intelligence must operate in real time, with limited power budgets, minimal buffering, and near-zero tolerance for failure. Memory must be local, fast, and reliable. Power delivery must be precise. The margin for error collapses.

As inference spreads outward to robots, vehicles, industrial systems, sensors, and autonomous platforms, the system increasingly rewards architectures and materials that perform under continuous stress rather than peak benchmarks. Reliability replaces raw throughput as the defining metric.

As memory became strategic, so too did the physical materials that enable fast, reliable signal transmission. In that sense, silver is not merely an industrial input, it is a scaling constraint for embodied intelligence, linking digital cognition to the physical world it increasingly inhabits.

Electricity as the Physical Analog of Memory

The physical world is now undergoing a parallel transition and its constraints are already visible.

The global economy is not short of energy. Fossil fuels remain abundant. Renewable capacity continues to expand. From a purely volumetric perspective, there is no immediate energy scarcity. But AI systems do not run on energy in the abstract. They run on electricity delivered with extreme precision, density, and reliability and that distinction is no longer academic.

Across the data-center ecosystem, shortages have already emerged in the infrastructure required to convert energy into usable electricity. Gas turbines face multi-year lead times. Transformers are constrained by manufacturing bottlenecks and grid backlogs. Switching gear, high-voltage components, and cooling systems are increasingly rate-limiting factors. In many regions, the constraint is no longer compute or capital, but whether power can be conditioned, delivered, and cooled fast enough to support new AI loads.

Timing matters. Voltage stability matters. Heat dissipation matters. Conversion losses matter. Failure tolerance is approaching zero.

Electricity plays the same role in the physical world that memory plays in the digital one. It determines whether theoretical capacity can be turned into real performance. Energy without electricity is like compute without memory: abundant, but unusable.

Silver sits precisely at this interface.

Silver as an Interface Material

Silver is not an energy commodity. It does not generate power. Instead, it appears at the narrow waist of electrical systems where electrons are forced through small spaces under high current, high frequency, and high reliability requirements.

It is used in contacts, switches, connectors, solders, conductive pastes, power electronics, and photovoltaic metallization. These are not headline components. They are failure points.

What matters about failure points is not volume, but system behavior. As electrical systems scale, performance does not degrade smoothly it fails discretely. Connectors trip. Switches arc. Data centers go offline. In exponential systems, failures propagate outward from interfaces. They do not average out; they cascade.

This is why silver demand rises as systems scale not because more silver is consumed in bulk, but because reliability margins collapse.

Synchronized S-Curves and Structural Scarcity

Solar power provides the clearest example. Despite a 95% reduction in silver intensity per watt since 2010, from roughly 400mg to just 13mg for modern N-type cells, total solar consumption has quadrupled, from 50 million ounces to 232 million ounces in 2024, as deployment overwhelmed efficiency gains (Silver Institute, World Silver Survey 2025). Despite years of “thrifting” silver per photovoltaic cell, total silver demand continues to rise because deployment has exploded. Solar scales by area, not device count, and utility-scale installations consume silver by the ton. As AI drives electricity demand higher, solar becomes one of the fastest and most politically viable ways to add generation capacity binding silver directly to AI’s energy supply chain.

Generation, however, is only the first layer.

Power must then be converted, conditioned, and distributed. Power electronics, grids, switchgear, and protection systems become stressed. These are precisely the junctions where silver’s conductivity and corrosion resistance matter most. At the device level, the same logic persists across data centers, edge devices, electric vehicles, autonomous systems, and military platforms.

Individually, none of these transitions would strain global silver supply. Together, they create a layered demand structure unlike anything seen before. Solar deployment, AI data centers, grid hardening, edge intelligence, electrified transport, industrial automation, and military modernization are all scaling simultaneously driven by different incentives but converging on the same failure points.

Supply does not respond symmetrically.

Most silver production is a byproduct of mining other metals, making supply structurally price inelastic. Recycling helps only at the margin. Substitution faces its own physics: alternatives like copper or aluminum degrade faster under high-frequency switching and corrosive conditions, making them unsuitable precisely where reliability margins are tightest. Unlike software, compute, or even energy generation, silver supply cannot scale exponentially.

This is not a temporary imbalance. It is structural and this is at the beginning of the Huang 15 year, “\$85 trillion dollar” infrastructure buildout.

The Bottleneck Revealed

This is the deeper pattern at work.

Exponential systems do not fail because they lack ideas. They fail because they concentrate stress. What begins as a software problem becomes a hardware problem. What begins as a hardware problem becomes a power problem. What

begins as a power problem becomes a materials problem. At each stage, abundance gives way to access, scale gives way to interfaces, and creativity gives way to physics.

AI is now far enough along this curve that its constraints have migrated downward out of models and into memory, electricity, and materials. DDR3 revealed what happens when production exits too early. High-bandwidth memory revealed what happens when access lags capability. Silver reveals what happens when strategic importance is misunderstood for too long.

This does not mean silver is the story of AI. It means silver is a signal a visible repricing of an invisible constraint. Exponential innovation rarely announces its bottlenecks in advance. They reveal themselves through stress, repricing, and the failure of substitution.

AI is not escaping the physical world. It is colliding with it.

And when exponential systems collide with physics, the future is shaped not by the largest components, but by the smallest ones that cannot fail.

Markets are already signaling to investors that the physical upgrade is underway. On the one hand, we have the parabolic rising of DRAM and silver prices. On the other hand, software is feeling constant pressure from rapid advances in coding automation and agentic AI. Year to date, the Level 2 GICS industry performance for the S&P 1500 shows Materials and Energy as the two best-performing groups, each up more than 9% through Friday, while Software is the worst performer, down more than 5%. This marks a notable inflection after more than 15 years in which software outperformed both energy and materials by over 1,400% since the Global Financial Crisis. As AI accelerates, I increasingly frame investing around scarcity rather than abundance, owning what becomes constrained while reducing exposure to what becomes commoditized. If this shift is real and now, silver and memory are not isolated anomalies but early signals of a broader repricing that will unfold as the estimated \$85 trillion physical-world upgrade progresses and AI progress and physics collide in 2026.

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